

BLOOD DONOR FINDER

“Every Drop Counts, Every Life Matters.”

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Document status: Proposed project design. Technology choices shown with "/" are alternatives to be finalized by the team. Features described as proposed should not be presented as already implemented until they are built and tested.

1. Project Overview

Blood Donor Finder is a proposed web-based platform that helps people search for potential blood donors using blood group and location. It is intended to make donor discovery more organized, especially when a blood request is urgent. A requester can select the required blood group and share a location through the browser's geolocation feature or enter a location manually. The system can then find matching registered donors and arrange the results by approximate distance.

A key feature of the proposal is automatic expanding-radius search. The application begins with a smaller search area and gradually expands it when no matching donors are found. This avoids asking the requester to repeatedly change the search distance themselves.

The system also checks the **nearest blood banks that have the required blood group available**, so the requester has another option besides individual donors.

The platform is intended to support communication between requesters and potential donors. It does not replace medical professionals, hospitals, blood banks, blood screening, or clinical decisions. A displayed match is only a potential lead; availability and eligibility must be confirmed through appropriate medical channels.

Project at a glance

Item	Description
Project name	Blood Donor Finder
Slogan	Every Drop Counts, Every Life Matters.
Project type	Location-based web application
Primary users	Blood donors, blood requesters and administrator
Main capability	Blood-group matching with location-based search
Highlighted features	Automatic expanding-radius search; nearest blood bank check by available blood group

2. Problem Statement

During a medical emergency, families may struggle to find a suitable blood donor quickly. They may have to contact several people, post requests in different groups, or search across multiple sources. These steps can take time, and the information found may be incomplete or out of date.

The main problem this project addresses is the absence of a simple, organized way to discover potential blood donors and nearby blood banks by blood group and approximate location.

Common difficulties include:

- Finding people who have registered with the required blood group.
- Identifying potential donors who are geographically nearby.
- Knowing whether a donor is currently willing and available to respond.
- Avoiding repeated manual searches across different distances.
- Communicating an urgent request without exposing unnecessary personal information.
- Finding which nearby blood banks have the required blood group available.

Proposed solution

Blood Donor Finder aims to provide one place to submit a request, search registered donor profiles, sort matching results by distance, check the nearest blood banks that have the required blood group, and coordinate notifications with the consent of donors. If no matches are found within the configured maximum radius, the system can retain the request and show alternative next steps.

3. Objectives and Scope

Project objectives

- Provide a straightforward registration process for donors.
- Allow requesters to search using a required blood group and location.
- Find potential matches within a configurable geographical radius.
- Expand the search area automatically when closer matches are unavailable.
- Display matching results in nearest-first order with approximate distances.
- Check the nearest blood banks that have the required blood group available.
- Support consent-based emergency notifications.
- Protect user accounts, contact details and location information.

Project scope

Included in proposed scope	Outside the system's role
Donor registration and profile management	Diagnosing a patient's medical condition
Blood group and location-based search	Determining clinical donor eligibility
Nearest blood bank check by blood group availability	Issuing or reserving blood units from a blood bank
Approximate distance and availability display	Guaranteeing that a donor will donate
Blood request creation and notifications	Replacing hospitals or licensed blood banks
Administrative review and basic verification	Performing blood testing or transfusion

Important: The application should describe search results as “potential matching donors”. Blood group matching alone is not sufficient to confirm safe or medically suitable donation. Blood bank availability is indicative and must be confirmed with the blood bank.

4. Proposed System and User Roles

The application is designed around three user roles.

4.1 Donor

- Create an account and provide required profile details.
- Enter blood group and location, subject to verification procedures.
- Update availability, pause notifications or deactivate the profile.
- Choose whether to receive emergency request notifications.
- Respond to a request using the contact method provided by the platform.

4.2 Blood requester

- Create or submit a blood request with the required blood group.
- Share current location through browser geolocation or enter a city/area manually.
- View potential matching donors sorted by approximate distance.
- View the nearest blood banks that have the required blood group available.
- See the radius in which results were found.
- Send or respond to a request through the platform's permitted communication flow.

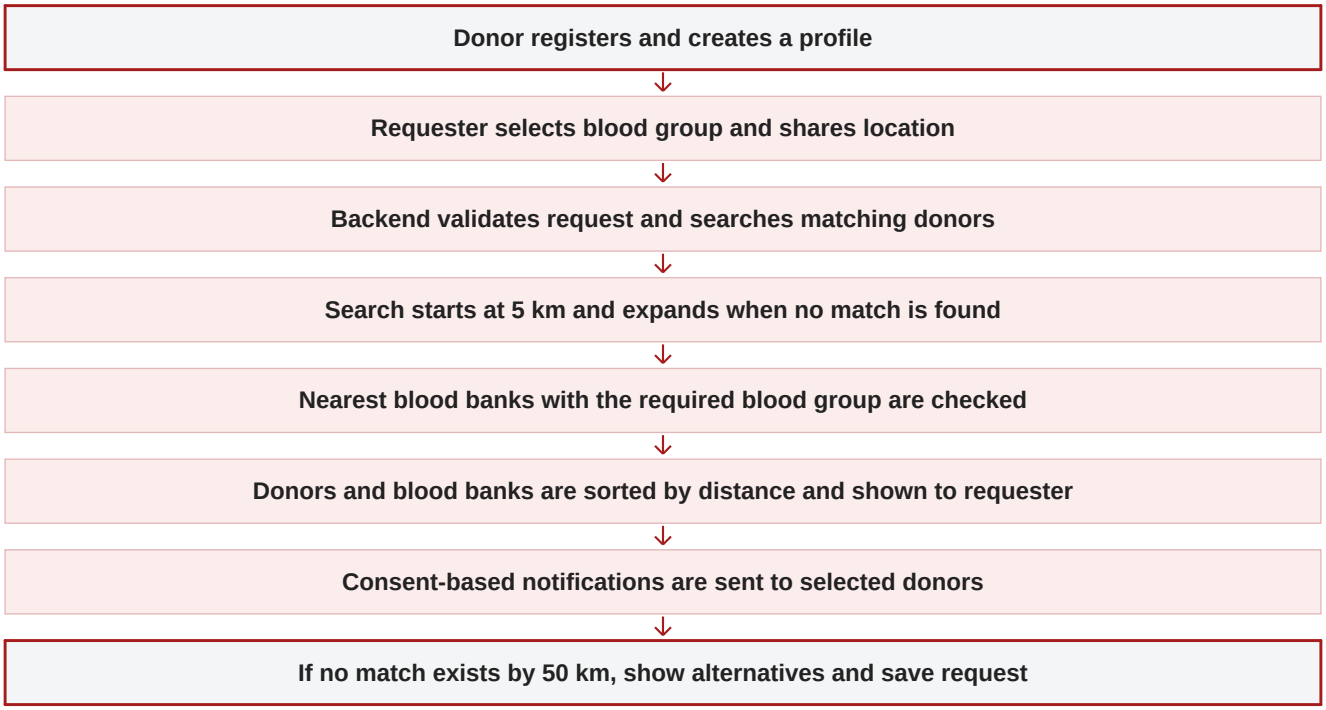
4.3 Administrator

- Review reported profiles and suspicious activity.
- Manage inappropriate or outdated listings according to platform policy.
- Maintain and verify blood bank listings and their blood group availability details.
- Monitor request status and notification failures.

- Maintain basic platform records and operational settings.

5. System Workflow

The following is the proposed end-to-end workflow, from donor registration to handling a request.



Workflow notes

- A donor profile should have an availability status that can be updated by the donor.
- The requester should be told how recently donor availability was updated, where appropriate.
- Blood bank results should show when their stock information was last updated.
- The application should not expose a donor's exact home coordinates to requesters.
- A request should have a status, such as Open, Responses Received, Fulfilled or Closed.

6. Automatic Expanding-Radius Search and Blood Bank Check

This feature reduces manual effort during a search. The requester does not need to increase the distance repeatedly. The backend performs the search using a predefined sequence of radius limits.

Search stage	Radius	System action
1	5 km	Search the closest area first.
2	10 km	Search a wider area if stage 1 has no matches.
3	20 km	Expand the search if no matches are found.
4	30 km	Continue searching a larger area.
5	50 km	Reach the configured maximum search radius.

Nearest blood bank check

Along with donor search, the system also checks the nearest blood banks that have the requested blood group available. This check uses the same location as the donor search and runs alongside it, so the requester sees blood bank options even while donors are still being searched.

- Blood banks that list the required blood group are shown nearest-first with approximate distance and contact details.
- Each result shows the blood groups listed as available and when this information was last updated.
- If no blood bank within the maximum radius lists the required blood group, the nearest blood banks are still shown with advice to contact them directly.
- Availability shown is indicative. The requester must confirm stock with the blood bank, and blood is issued only by a licensed blood bank under its own rules.

7. Technology Stack

The following stack is taken from the team's provided technology plan. Items separated by a slash are alternatives; the team should select one option for each layer before implementation.

Layer	Technology	How it is used
Frontend	HTML, CSS, JavaScript, React.js, Tailwind CSS / Bootstrap	Build responsive pages, forms and interactive search results.
Backend	Python (Flask / FastAPI)	Implement APIs, authentication flows, donor search and request handling.
Database	MySQL / MongoDB	Store donor profiles, blood requests, user accounts, blood bank records and notification records.
Authentication	JWT, bcrypt	Manage authenticated sessions/tokens and securely hash passwords.
Maps and location	Leaflet.js, OpenStreetMap, Browser Geolocation API	Collect requester location with permission and display map information.
Distance calculation	Haversine formula / geospatial queries	Calculate or filter donors and blood banks by distance.
Blood bank data	Blood bank records in the database; optional official directory such as e-RaktKosh (subject to access permission)	Store blood bank name, location, contact and blood group availability for the nearest blood bank check.
Notifications	Fast2SMS or Twilio (SMS); Nodemailer or SendGrid (Email)	Send request notifications and relevant updates.
Tools	Git, GitHub, Postman, VS Code	Version control, collaboration, API testing and development.
Deployment	Vercel / Netlify (frontend); Render / Railway (backend); MongoDB Atlas (database)	Host the application and managed database where applicable.

Implementation note: MongoDB Atlas applies only if MongoDB is selected. If the team chooses MySQL, use a suitable MySQL hosting service instead. Likewise, Flask and FastAPI are alternative backend frameworks, not technologies that must both be used.

8. High-Level System Architecture

The application can be organized into four main layers. The browser communicates with the backend through API requests. The backend validates each request, applies business rules, reads or updates the database, and calls external services when required.

Layer	Responsibilities
User interface	React-based donor and requester pages, forms, search results and map view.
Backend API	Registration, login, validation, donor matching, blood bank lookup, request management and notification orchestration.

Layer	Responsibilities
Database	User and donor profiles, location data, blood bank records, requests, consent preferences and notification status.
External services	Map tiles/geocoding as configured, optional blood bank directory, SMS provider and email provider.

Typical request sequence

- The requester submits a blood group and location from the web interface.
- The frontend sends the search request to the backend API.
- The backend validates the input and searches the database for matching donors.
- The backend calculates or retrieves distances and applies the radius rules.
- The backend also looks up the nearest blood banks that list the required blood group.
- The backend returns donor and blood bank results with approximate distances and permitted display details.
- If an emergency request is submitted, the notification service contacts eligible, opted-in donors according to the configured stages.

This is a logical architecture. The final implementation may combine or separate services depending on the selected framework and hosting setup.

9. Core Data Model

The following entities/collections are a starting point for database design. The exact fields should be adjusted to the selected database and the team's implementation.

Entity	Suggested fields
User	user_id, name, email/phone, password_hash, role, created_at, account_status
DonorProfile	donor_id, user_id, blood_group, availability_status, location, location_updated_at, verification_status, notification_consent
BloodBank	bank_id, name, address, location, contact, license_details, available_blood_groups, stock_updated_at
BloodRequest	request_id, requester_id, required_blood_group, request_location, urgency, units_requested, hospital/area, status, created_at
Notification	notification_id, request_id, donor_id, channel, sent_at, delivery_status, response_status
Admin / Audit	admin_id, user_id, action, target_record, timestamp

Location representation

For GeoJSON-based storage, a location can be represented as a Point. GeoJSON coordinates use longitude first, then latitude.

```
{ "type": "Point", "coordinates": [longitude, latitude] }
```

A donor location should only be collected with permission and should be stored and displayed at the level needed for matching. Avoid exposing exact coordinates or a home address in public search results.

10. Security, Privacy and Responsible Use

Because the platform handles contact details and location information, privacy and security must be considered from the beginning.

- **Password security:** Store password hashes using bcrypt. Never store plain-text passwords.
- **Authentication:** Use JWT carefully, set appropriate expiry, and protect secret keys outside source code.
- **Input validation:** Validate blood group, location coordinates, phone/email fields and request data on the backend.

- **Location privacy:** Ask for browser location permission, offer manual entry, and show approximate distance rather than exact donor coordinates.
- **Donor consent:** Let donors choose whether to receive emergency alerts and how their contact details may be shared.
- **Availability:** Allow donors to pause alerts and update availability. A profile should not imply that a donor is currently ready to donate unless they have confirmed it.
- **Blood bank information:** Show only verified contact details and clearly mark blood group availability as indicative with its last-updated time.
- **Data retention:** Define how long inactive profiles, closed requests and notification records are retained.
- **Abuse prevention:** Add rate limits, reporting, admin review and safeguards against repeated or fraudulent requests.

Medical and operational disclaimer
Blood Donor Finder is an information and coordination tool. It does not verify medical suitability, perform blood testing, reserve blood units, or guarantee donation or blood availability. Requesters should contact hospitals or licensed blood banks promptly and follow professional medical guidance. The platform should avoid presenting itself as an emergency medical service.

11. Advantages

Less manual effort
The requester does not need to repeatedly change the search radius.

Location-aware results
Potential donors can be ordered by approximate distance.

Wider search when needed
The system can expand the search area when a smaller area has no matches.

Nearest blood bank option
Requesters can also see the nearest blood banks that have the required blood group available.

Organized information
Donor profiles, blood banks and blood requests can be maintained in one platform.

Structured communication
Consent-based notifications can help requesters reach relevant donors.

Clearer feedback
Messages such as “Matching donors found within 10 km” explain the search result.

Room for expansion
The platform can later integrate live blood bank stock, mobile apps and additional notification channels.

12. Limitations and Challenges

Challenge	Why it matters	Possible mitigation
Outdated donor availability	A listed donor may no longer be available.	Let donors update status and show last-updated time.
Blood bank stock accuracy	Blood stock changes quickly and listed data may be outdated.	Show last-updated time and ask users to confirm availability by phone.
Location accuracy	GPS may be unavailable or imprecise.	Allow manual location entry and explain approximate distances.
Privacy risk	Contact and location details are personal information.	Use consent, access controls and approximate location display.
Notification delays	SMS/email delivery is not guaranteed.	Track delivery status and show other contact/medical options.
False or duplicate profiles	Unreliable records can waste time.	Verify contact details and provide reporting/admin review.
Long travel distance	A donor far away may not be practical for an urgent request.	Show distance clearly and direct users to nearby hospitals/blood banks.
Technical complexity	Maps, authentication and geospatial search require testing.	Build in stages and test each module independently.

The system should communicate uncertainty honestly. A search result means that a profile or blood bank appears to match the request criteria; it does not confirm availability, medical eligibility or successful donation.

13. Testing Plan

The team should test normal cases, edge cases and failure cases before presenting the system as complete.

Test case	Expected result
Donor registration with valid details	Profile is created and stored successfully.
Registration with invalid or missing fields	Backend returns clear validation errors.
Login with incorrect password	Access is denied without revealing sensitive account information.
Search with a valid blood group and location	Matching profiles are returned, sorted by distance.
No matches within 5 km but matches within 10 km	Search expands and reports the successful radius.
No matches within 50 km	System displays no-match guidance and saves the request if submitted.
Blood bank lists the requested blood group	Blood bank is shown with distance, contact and last-updated time.
No nearby blood bank lists the requested blood group	Nearest blood banks are shown with advice to contact them directly.
Location permission denied	User can enter a location manually.
Donor has opted out of notifications	No emergency alert is sent to that donor.
SMS/email provider fails	Failure is recorded and the user is not falsely told it was delivered.
Donor marks unavailable	Donor is excluded from available-donor results or clearly labelled unavailable.

14. Future Scope

- **Mobile application:** Build Android and iOS versions for easier access.
- **Live blood bank stock integration:** Connect with official blood bank systems for more reliable, real-time availability.
- **Map-based search:** Add a map view with privacy-safe approximate donor markers and blood bank locations.
- **Multilingual interface:** Support regional languages to make the platform easier to use.
- **Improved verification:** Introduce stronger profile verification and reporting workflows.
- **Request tracking:** Allow requesters to track responses and close fulfilled or expired requests.
- **Analytics dashboard:** Provide aggregate statistics for administrators without exposing personal data.
- **Smarter notification rules:** Configure distance bands, waiting periods and donor response handling.

Future features should be added only after the core registration, search, privacy and request-management workflows are reliable.

15. Conclusion

Blood Donor Finder is a proposed web application that aims to make the discovery of potential blood donors more organized and location-aware. By combining blood group selection, geographical search, automatic radius expansion, nearest blood bank check and consent-based notifications, the project addresses several practical difficulties that people may face while looking for blood.

The automatic expanding-radius feature is a central part of the proposal: the system starts with nearby areas and widens the search when no matching profiles are found, while also pointing the requester to nearby blood banks that have the required blood group. Clear distance information, donor availability controls and privacy safeguards can make this process more useful and responsible.

The project is also an opportunity for the team to apply frontend development, backend APIs, database design, authentication, geolocation, distance calculations and software testing to a meaningful real-world problem. With careful implementation and appropriate coordination with hospitals and blood banks, the platform can become a helpful support tool for blood-related requests.

16. Legal Notice

Sale and purchase of blood is illegal in India. Human blood is regulated as a “drug” under the Drugs and Cosmetics Act, 1940 and the Drugs and Cosmetics Rules, 1945. Following the Supreme Court's 1996 judgment in *Common Cause v. Union of India*, paid and professional blood donation has been banned since 1998. Blood must be donated voluntarily and without payment.

Blood Donor Finder is a free coordination platform. Users must not offer, request, advertise or accept money or any other reward for blood. Such listings or requests may be removed, and the accounts concerned may be blocked or reported to the authorities. Any charges made by licensed blood banks relate to processing and testing, not to the sale of blood.

This notice is general information and not legal advice.

Every Drop Counts, Every Life Matters.

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References for implementation

- Python documentation — https://docs.python.org/
- Flask documentation — https://flask.palletsprojects.com/ or FastAPI documentation — https://fastapi.tiangolo.com/
- MongoDB geospatial queries — https://www.mongodb.com/docs/manual/geospatial-queries/
- MySQL spatial reference — https://dev.mysql.com/doc/
- Leaflet documentation — https://leafletjs.com/reference.html
- OpenStreetMap — https://www.openstreetmap.org/